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Quiz 8

Difference between Stochastic vs Deterministic Policy Gradient:

Aspect	Stochastic Policy Gradient	Deterministic Policy Gradient
Policy expression	$\pi_\theta(a s)$ (probability distribution)	$\mu_\theta(s)$ (deterministic mapping)
Action output	Sample from distribution	Direct deterministic action
Gradient formula	$\nabla_\theta J = \mathbb{E}_{s,a} [\nabla_\theta \log \pi_\theta(a s) Q^\pi(s, a)]$	$\nabla_\theta J = \mathbb{E}_s [\nabla_a Q^\mu(s, a) _{a=\mu_\theta(s)} \nabla_\theta \mu_\theta(s)]$
Expectation scope	State & action	Only state
Exploration	Built-in by sampling	Need artificial noise
Variance	Higher	Lower
Typical algorithms	REINFORCE, Actor-Critic, PPO	DDPG
Action space	Discrete & continuous	Mainly continuous